2.5.1 Matrix Representation of Kets, Bras, and Operators

Consider a discrete, complete, and orthonormal basis which is made of an infinite[4] set of kets $|\,\phi_1\rangle, |\,\phi_2\rangle, |\,\phi_3\rangle, \ldots, |\,\phi_n\rangle$ and denote it by $\{|\,\phi_n\rangle\}$. Note that the basis $\{|\,\phi_n\rangle\}$ is discrete, yet it has an infinite number of unit vectors. In the limit $n \to \infty$, the ordering index n of the unit vectors $|\,\phi_n\rangle$ is *discrete* or *countable*; that is, the sequence $|\,\phi_1\rangle, |\,\phi_2\rangle, |\,\phi_3\rangle, \ldots$ is countably infinite. As an illustration, consider the special functions, such as the Hermite, Legendre, or Laguerre polynomials, $H_n(x)$, $P_n(x)$, and $L_n(x)$. These polynomials are identified by a discrete index n and by a continuous variable x; although n varies discretely, it can be infinite.

In Section 2.6, we will consider bases that have a continuous and infinite number of base vectors; in these bases the index n increases continuously. Thus, each basis has a continuum of base vectors.

In this section the notation $\{|\,\phi_n\rangle\}$ will be used to abbreviate an infinitely countable set of vectors (i.e., $|\,\phi_1\rangle, |\,\phi_2\rangle, |\,\phi_3\rangle, \ldots$) of the Hilbert space $\mathcal{H}$. The orthonormality condition of the base kets is expressed by

$$\langle\phi_n \mid \phi_m\rangle = \delta_{nm}, \tag{2.170}$$

where δ_{nm} is the *Kronecker delta* symbol defined by

$$\delta_{nm} = \begin{cases} 1, & n = m, \\ 0, & n \neq m. \end{cases} \quad . \tag{2.171}$$

The completeness, or closure, relation for this basis is given by

$$\sum_{n=1}^{\infty} |\,\phi_n\rangle\langle\phi_n\,| = \hat{I}, \tag{2.172}$$

where $\hat{I}$ is the unit operator; when the unit operator acts on any ket, it leaves the ket unchanged.

2.5.1.1 Matrix Representation of Kets and Bras

Let us now examine how to represent the vector $|\,\psi\rangle$ within the context of the basis $\{|\,\phi_n\rangle\}$. The completeness property of this basis enables us to expand any state vector $|\,\psi\rangle$ in terms of the base kets $|\,\phi_n\rangle$:

$$|\,\psi\rangle = \hat{I}\,|\,\psi\rangle = \left(\sum_{n=1}^{\infty} |\,\phi_n\rangle\langle\phi_n\,|\right)|\,\psi\rangle = \sum_{n=1}^{\infty} a_n\,|\,\phi_n\rangle, \tag{2.173}$$

where the coefficient a_n, which is equal to $\langle\phi_n \mid \psi\rangle$, represents the projection of $|\,\psi\rangle$ onto $|\,\phi_n\rangle$; a_n is the component of $|\,\psi\rangle$ along the vector $|\,\phi_n\rangle$. Recall that the coefficients a_n are complex numbers. So, within the basis $\{|\,\phi_n\rangle\}$, the ket $|\,\psi\rangle$ is represented by the set of its components, $a_1, a_2, a_3, \ldots$ along $|\,\phi_1\rangle, |\,\phi_2\rangle, |\,\phi_3\rangle, \ldots$, respectively. Hence $|\,\psi\rangle$ can be represented by a *column* vector which has a countably infinite number of components:

$$|\,\psi\rangle \longrightarrow \begin{pmatrix} \langle\phi_1 \mid \psi\rangle \\ \langle\phi_2 \mid \psi\rangle \\ \vdots \\ \langle\phi_n \mid \psi\rangle \\ \vdots \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \\ \vdots \end{pmatrix}. \tag{2.174}$$

[4] Kets are elements of the Hilbert space, and the dimension of a Hilbert space is infinite.

The bra $\langle\psi\mid$ can be represented by a *row* vector:

$$\begin{aligned}\langle\psi\mid \longrightarrow\ & (\langle\psi\mid\phi_1\rangle\ \langle\psi\mid\phi_2\rangle\ \cdots\ \langle\psi\mid\phi_n\rangle\ \cdots)\\ =\ & (\langle\phi_1\mid\psi\rangle^*\ \langle\phi_2\mid\psi\rangle^*\ \cdots\ \langle\phi_n\mid\psi\rangle^*\ \cdots)\\ =\ & (a_1^*\ a_2^*\ \cdots\ a_n^*\ \cdots).\end{aligned} \tag{2.175}$$

Using this representation, we see that a bra-ket $\langle\psi\mid\phi\rangle$ is a complex number equal to the matrix product of the row matrix corresponding to the bra $\langle\psi\mid$ with the column matrix corresponding to the ket $\mid\phi\rangle$:

$$\langle\psi\mid\phi\rangle = (a_1^*\ a_2^*\ \cdots\ a_n^*\ \cdots)\begin{pmatrix} b_1\\ b_2\\ \vdots\\ b_n\\ \vdots\end{pmatrix} = \sum_n a_n^* b_n, \tag{2.176}$$

where $b_n = \langle\phi_n\mid\phi\rangle$. We see that, within this representation, the matrices representing $\mid\psi\rangle$ and $\langle\psi\mid$ are Hermitian adjoints of each other.

Remark

A ket $|\psi\rangle$ is normalized if $\langle\psi\mid\psi\rangle = \sum_n |a_n|^2 = 1$. If $|\psi\rangle$ is not normalized and we want to normalized it, we need simply to multiply it by a constant α so that $\langle\alpha\psi\mid\alpha\psi\rangle = |\alpha|^2\langle\psi\mid\psi\rangle = 1$, and hence $\alpha = 1/\sqrt{\langle\psi\mid\psi\rangle}$.

Example 2.12

Consider the following two kets:

$$\mid\psi\rangle = \begin{pmatrix} 5i\\ 2\\ -i\end{pmatrix}, \qquad \mid\phi\rangle = \begin{pmatrix} 3\\ 8i\\ -9i\end{pmatrix}.$$

(a) Find $\mid\psi\rangle^*$ and $\langle\psi\mid$.
(b) Is $\mid\psi\rangle$ normalized? If not, normalize it.
(c) Are $\mid\psi\rangle$ and $\mid\phi\rangle$ orthogonal?

Solution

(a) The expressions of $\mid\psi\rangle^*$ and $\langle\psi\mid$ are given by

$$\mid\psi\rangle^* = \begin{pmatrix} -5i\\ 2\\ i\end{pmatrix}, \qquad \langle\psi\mid\ =\ (-5i\quad 2\quad i)\,, \tag{2.177}$$

where we have used the fact that $\langle\psi\mid$ is equal to the complex conjugate of the transpose of the ket $\mid\psi\rangle$. Hence, we should reiterate the important fact that $\mid\psi\rangle^* \neq \langle\psi\mid$.

(b) The norm of $\mid\psi\rangle$ is given by

$$\langle\psi\mid\psi\rangle = (-5i\quad 2\quad i)\begin{pmatrix} 5i\\ 2\\ -i\end{pmatrix} = (-5i)(5i) + (2)(2) + (i)(-i) = 30. \tag{2.178}$$

Thus, $|\,\psi\rangle$ is not normalized. By multiplying it with $1/\sqrt{30}$, it becomes normalized:

$$|\,\chi\rangle = \frac{1}{\sqrt{30}}\,|\,\psi\rangle = \frac{1}{\sqrt{30}}\begin{pmatrix} 5i \\ 2 \\ -i \end{pmatrix} \qquad \Longrightarrow \qquad \langle\chi\,|\,\chi\rangle = 1. \tag{2.179}$$

(c) The kets $|\,\psi\rangle$ and $|\,\phi\rangle$ are not orthogonal since their scalar product is not zero:

$$\langle\psi\,|\,\phi\rangle = (-5i \quad 2 \quad i)\begin{pmatrix} 3 \\ 8i \\ -9i \end{pmatrix} = (-5i)(3) + (2)(8i) + (i)(-9i) = 9 + i. \tag{2.180}$$

2.5.1.2 Matrix Representation of Operators

For each linear operator $\hat{A}$, we can write

$$\hat{A} = \hat{I}\hat{A}\hat{I} = \left(\sum_{n=1}^{\infty} |\,\phi_n\rangle\langle\phi_n\,|\right)\hat{A}\left(\sum_{m=1}^{\infty} |\,\phi_m\rangle\langle\phi_m\,|\right) = \sum_{nm} A_{nm}\,|\,\phi_n\rangle\langle\phi_m\,|, \tag{2.181}$$

where A_{nm} is the nm matrix element of the operator $\hat{A}$:

$$A_{nm} \;=\; \langle\phi_n\,|\,\hat{A}\,|\,\phi_m\rangle. \tag{2.182}$$

We see that the operator $\hat{A}$ is represented, within the basis $\{|\,\phi_n\rangle\}$, by a *square* matrix A (A without a hat designates a matrix), which has a countably infinite number of columns and a countably infinite number of rows:

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ A_{31} & A_{32} & A_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \tag{2.183}$$

For instance, the *unit* operator $\hat{I}$ is represented by the unit matrix; when the unit matrix is multiplied with another matrix, it leaves that unchanged:

$$I = \begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{2.184}$$

In summary, *kets are represented by column vectors, bras by row vectors, and operators by square matrices*.

2.5.1.3 Matrix Representation of Some Other Operators

(a) Hermitian adjoint operation

Let us now look at the matrix representation of the Hermitian adjoint operation of an operator. First, recall that the *transpose* of a matrix A, denoted by A^T, is obtained by interchanging the rows with the columns:

$$(A^T)_{nm} = A_{mn} \quad \text{or} \quad \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ A_{31} & A_{32} & A_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}^T = \begin{pmatrix} A_{11} & A_{21} & A_{31} & \cdots \\ A_{12} & A_{22} & A_{32} & \cdots \\ A_{13} & A_{23} & A_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{2.185}$$

Similarly, the transpose of a column matrix is a row matrix, and the transpose of a row matrix is a column matrix:

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \\ \vdots \end{pmatrix}^T = \begin{pmatrix} a_1 & a_2 & \cdots & a_n & \cdots \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} a_1 & a_2 & \cdots & a_n & \cdots \end{pmatrix}^T = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \\ \vdots \end{pmatrix}. \tag{2.186}$$

So a square matrix A is symmetric if it is equal to its transpose, $A^T = A$. A skew-symmetric matrix is a square matrix whose transpose equals the negative of the matrix, $A^T = -A$.

The *complex conjugate* of a matrix is obtained by simply taking the complex conjugate of all its elements: $(A^*)_{nm} = (A_{nm})^*$.

The matrix which represents the operator $\hat{A}^\dagger$ is obtained by taking the complex conjugate of the matrix transpose of A:

$$A^\dagger = (A^T)^* \quad \text{or} \quad (\hat{A}^\dagger)_{nm} = \langle \phi_n \mid \hat{A}^\dagger \mid \phi_m \rangle = \langle \phi_m \mid \hat{A} \mid \phi_n \rangle^* = A^*_{mn}; \tag{2.187}$$

that is,

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ A_{31} & A_{32} & A_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}^\dagger = \begin{pmatrix} A^*_{11} & A^*_{21} & A^*_{31} & \cdots \\ A^*_{12} & A^*_{22} & A^*_{32} & \cdots \\ A^*_{13} & A^*_{23} & A^*_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{2.188}$$

If an operator $\hat{A}$ is Hermitian, its matrix satisfies this condition:

$$(A^T)^* = A \quad \text{or} \quad A^*_{mn} = A_{nm}. \tag{2.189}$$

The diagonal elements of a Hermitian matrix therefore must be real numbers. Note that *a Hermitian matrix must be square*.

(b) Inverse and unitary operators

A matrix has an inverse only if it is square and its determinant is nonzero; a matrix that has an inverse is called a nonsingular matrix and a matrix that has no inverse is called a singular

matrix. The elements A^{-1}_{nm} of the inverse matrix A^{-1}, representing an operator $\hat{A}^{-1}$, are given by the relation

$$A^{-1}_{nm} = \frac{\text{cofactor of } A_{mn}}{\text{determinant of } A} \quad \text{or} \quad A^{-1} = \frac{B^T}{\text{determinant of } A}, \tag{2.190}$$

where B is the matrix of *cofactors* (also called the minor); the cofactor of element A_{mn} is equal to $(-1)^{m+n}$ times the determinant of the submatrix obtained from A by removing the mth row and the nth column. Note that when the matrix, representing an operator, has a determinant equal to zero, this operator does not possess an inverse. Note that $A^{-1}A = AA^{-1} = I$ where I is the unit matrix.

The inverse of a product of matrices is obtained as follows:

$$(ABC\cdots PQ)^{-1} = Q^{-1}P^{-1}\cdots C^{-1}B^{-1}A^{-1}. \tag{2.191}$$

The inverse of the inverse of a matrix is equal to the matrix itself, $\left(A^{-1}\right)^{-1} = A$.

A *unitary* operator $\hat{U}$ is represented by a unitary matrix. A matrix U is said to be unitary if its inverse is equal to its adjoint:

$$U^{-1} = U^{\dagger} \quad \text{or} \quad U^{\dagger}U = I, \tag{2.192}$$

where I is the unit matrix.

Example 2.13 (Inverse of a matrix)

Calculate the inverse of the matrix $A = \begin{pmatrix} 2 & i & 0 \\ 3 & 1 & 5 \\ 0 & -i & -2 \end{pmatrix}$. Is this matrix unitary?

Solution

Since the determinant of A is $\det(A) = -4 + 16i$, we have $A^{-1} = B^T/(-4+16i)$, where the elements of the cofactor matrix B are given by $B_{nm} = (-1)^{n+m}$ times the determinant of the submatrix obtained from A by removing the nth row and the mth column. In this way, we have

$$B_{11} = (-1)^{1+1}\begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} = (-1)^2\begin{vmatrix} 1 & 5 \\ -i & -2 \end{vmatrix} = -2 + 5i, \tag{2.193}$$

$$B_{12} = (-1)^{1+2}\begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} = (-1)^3\begin{vmatrix} 3 & 5 \\ 0 & -2 \end{vmatrix} = 6, \tag{2.194}$$

$$B_{13} = (-1)^{1+3}\begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{vmatrix} = (-1)^4\begin{vmatrix} 3 & 1 \\ 0 & -i \end{vmatrix} = -3i, \tag{2.195}$$

$$B_{21} = (-1)^3\begin{vmatrix} i & 0 \\ -i & -2 \end{vmatrix} = 2i, \qquad B_{22} = (-1)^4\begin{vmatrix} 2 & 0 \\ 0 & -2 \end{vmatrix} = -4, \tag{2.196}$$

$$B_{23} = (-1)^5\begin{vmatrix} 2 & i \\ 0 & -i \end{vmatrix} = 2i, \qquad B_{31} = (-1)^4\begin{vmatrix} i & 0 \\ 1 & 5 \end{vmatrix} = 5i, \tag{2.197}$$

$$B_{32} = (-1)^5\begin{vmatrix} 2 & 0 \\ 3 & 5 \end{vmatrix} = -10, \qquad B_{33} = (-1)^6\begin{vmatrix} 2 & i \\ 3 & 1 \end{vmatrix} = 2 - 3i, \tag{2.198}$$

and hence

$$B = \begin{pmatrix} -2+5i & 6 & -3i \\ 2i & -4 & 2i \\ 5i & -10 & 2-3i \end{pmatrix}. \tag{2.199}$$

Taking the transpose of B, we obtain

$$\begin{aligned} A^{-1} &= \frac{1}{-4+16i} B^T = \frac{-1-4i}{68} \begin{pmatrix} -2+5i & 2i & 5i \\ 6 & -4 & -10 \\ -3i & 2i & 2-3i \end{pmatrix} \\ &= \frac{1}{68} \begin{pmatrix} 22+3i & 8-2i & 20-5i \\ -6-24i & 4+16i & 10+40i \\ -12+3i & 8-2i & -14-5i \end{pmatrix}. \end{aligned} \tag{2.200}$$

Clearly, this matrix is not unitary since its inverse is not equal to its Hermitian adjoint: $A^{-1} \neq A^{\dagger}$.

(c) Matrix representation of $|\ \psi\rangle\langle\psi\ |$

It is now easy to see that the product $|\ \psi\rangle\langle\psi\ |$ is indeed an operator, since its representation within $\{|\ \phi_n\rangle\}$ is a square matrix:

$$|\ \psi\rangle\langle\psi\ | = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \end{pmatrix} (a_1^* \quad a_2^* \quad a_3^* \quad \cdots) = \begin{pmatrix} a_1 a_1^* & a_1 a_2^* & a_1 a_3^* & \cdots \\ a_2 a_1^* & a_2 a_2^* & a_2 a_3^* & \cdots \\ a_3 a_1^* & a_3 a_2^* & a_3 a_3^* & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{2.201}$$

(d) Trace of an operator

The trace $\mathrm{Tr}(\hat{A})$ of an operator $\hat{A}$ is given, within an orthonormal basis $\{|\ \phi_n\rangle\}$, by the expression

$$\mathrm{Tr}(\hat{A}) = \sum_n \langle \phi_n\ |\ \hat{A}\ |\ \phi_n\rangle\ = \sum_n A_{nn}; \tag{2.202}$$

we will see later that the trace of an operator does not depend on the basis. The trace of a matrix is equal to the sum of its diagonal elements:

$$\mathrm{Tr} \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ A_{31} & A_{32} & A_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = A_{11} + A_{22} + A_{33} + \cdots. \tag{2.203}$$

Properties of the trace

We can ascertain that

$$\mathrm{Tr}(\hat{A}^{\dagger}) = (\mathrm{Tr}(\hat{A}))^*, \tag{2.204}$$

$$\mathrm{Tr}(\alpha \hat{A} + \beta \hat{B} + \gamma \hat{C} + \cdots) = \alpha \mathrm{Tr}(\hat{A}) + \beta \mathrm{Tr}(\hat{B}) + \gamma\, \mathrm{Tr}(\hat{C}) + \cdots, \tag{2.205}$$

and the trace of a product of operators is invariant under the cyclic permutations of these operators:

$$\mathrm{Tr}(\hat{A}\hat{B}\hat{C}\hat{D}\hat{E}) = \mathrm{Tr}(\hat{E}\hat{A}\hat{B}\hat{C}\hat{D}) = \mathrm{Tr}(\hat{D}\hat{E}\hat{A}\hat{B}\hat{C}) = \mathrm{Tr}(\hat{C}\hat{D}\hat{E}\hat{A}\hat{B}) = \cdots. \tag{2.206}$$

Example 2.14

(a) Show that $\mathrm{Tr}(\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{A})$.

(b) Show that the trace of a commutator is always zero.

(c) Illustrate the results shown in (a) and (b) on the following matrices:

$$A = \begin{pmatrix} 8-2i & 4i & 0 \\ 1 & 0 & 1-i \\ -8 & i & 6i \end{pmatrix}, \qquad B = \begin{pmatrix} -i & 2 & 1-i \\ 6 & 1+i & 3i \\ 1 & 5+7i & 0 \end{pmatrix}.$$

Solution

(a) Using the definition of the trace,

$$\mathrm{Tr}(\hat{A}\hat{B}) = \sum_n \langle \phi_n \mid \hat{A}\hat{B} \mid \phi_n \rangle, \tag{2.207}$$

and inserting the unit operator between $\hat{A}$ and $\hat{B}$ we have

$$\begin{aligned} \mathrm{Tr}(\hat{A}\hat{B}) &= \sum_n \langle \phi_n \mid \hat{A} \left(\sum_m \mid \phi_m \rangle\langle \phi_m \mid \right) \hat{B} \mid \phi_n \rangle = \sum_{nm} \langle \phi_n \mid \hat{A} \mid \phi_m \rangle \langle \phi_m \mid \hat{B} \mid \phi_n \rangle \\ &= \sum_{nm} A_{nm} B_{mn}. \end{aligned} \tag{2.208}$$

On the other hand, since $\mathrm{Tr}(\hat{A}\hat{B}) = \sum_n \langle \phi_n \mid \hat{A}\hat{B} \mid \phi_n \rangle$, we have

$$\begin{aligned} \mathrm{Tr}(\hat{B}\hat{A}) &= \sum_m \langle \phi_m \mid \hat{B} \left(\sum_n \mid \phi_n \rangle\langle \phi_n \mid \right) \hat{A} \mid \phi_m \rangle = \sum_m \langle \phi_m \mid \hat{B} \mid \phi_n \rangle \langle \phi_n \mid \hat{A} \mid \phi_m \rangle \\ &= \sum_{nm} B_{mn} A_{nm}. \end{aligned} \tag{2.209}$$

Comparing (2.208) and (2.209), we see that $\mathrm{Tr}(\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{A})$.

(b) Since $\mathrm{Tr}(\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{A})$ we can infer at once that the trace of any commutator is always zero:

$$\mathrm{Tr}([\hat{A},\ \hat{B}]) = \mathrm{Tr}(\hat{A}\hat{B}) - \mathrm{Tr}(\hat{B}\hat{A}) = 0. \tag{2.210}$$

(c) Let us verify that the traces of the products AB and BA are equal. Since

$$AB = \begin{pmatrix} -2+16i & 12 & -6-10i \\ 1-2i & 14+2i & 1-i \\ 20i & -59+31i & -11+8i \end{pmatrix}, \quad BA = \begin{pmatrix} -8 & 5+i & 8+4i \\ 49-35i & -3+24i & -16 \\ 13+5i & 4i & 12+2i \end{pmatrix}, \tag{2.211}$$

we have

$$\mathrm{Tr}(AB) = \mathrm{Tr} \begin{pmatrix} -2+16i & 12 & -6-10i \\ 1-2i & 14+2i & 1-i \\ 20i & -59+31i & -11+8i \end{pmatrix} = 1+26i, \tag{2.212}$$

$$\mathrm{Tr}(BA) = \mathrm{Tr} \begin{pmatrix} -8 & 5+i & 8+4i \\ 49-35i & -3+24i & -16 \\ 13+5i & 4i & 12+2i \end{pmatrix} = 1+26i = \mathrm{Tr}(AB). \tag{2.213}$$

This leads to $\mathrm{Tr}(AB) - \mathrm{Tr}(BA) = (1+26i) - (1+26i) = 0$ or $\mathrm{Tr}([A,\ B]) = 0$.

2.5.1.4 Matrix Representation of Several Other Quantities

(a) Matrix representation of $|\,\phi\rangle = \hat{A}\,|\,\psi\rangle$
The relation $|\,\phi\rangle = \hat{A}\,|\,\psi\rangle$ can be cast into the algebraic form $\hat{I}\,|\,\phi\rangle = \hat{I}\hat{A}\hat{I}\,|\,\psi\rangle$ or

$$\left(\sum_n |\,\phi_n\rangle\langle\phi_n\,|\right)|\,\phi\rangle = \left(\sum_n |\,\phi_n\rangle\langle\phi_n\,|\right)\hat{A}\left(\sum_m |\,\phi_m\rangle\langle\phi_m\,|\right)|\,\psi\rangle, \tag{2.214}$$

which in turn can be written as

$$\sum_n b_n\,|\,\phi_n\rangle = \sum_{nm} a_m\,|\,\phi_n\rangle\langle\phi_n\,|\,\hat{A}\,|\,\phi_m\rangle = \sum_{nm} a_m A_{nm}\,|\,\phi_n\rangle, \tag{2.215}$$

where $b_n = \langle\phi_n\,|\,\phi\rangle$, $A_{nm} = \langle\phi_n\,|\,\hat{A}\,|\,\phi_m\rangle$, and $a_m = \langle\phi_m\,|\,\psi\rangle$. It is easy to see that (2.215) yields $b_n = \sum_m A_{nm}a_m$; hence the matrix representation of $|\,\phi\rangle = \hat{A}\,|\,\psi\rangle$ is given by

$$\begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ A_{31} & A_{32} & A_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \end{pmatrix}. \tag{2.216}$$

(b) Matrix representation of $\langle\phi\,|\,\hat{A}\,|\,\psi\rangle$
As for $\langle\phi\,|\,\hat{A}\,|\,\psi\rangle$ we have

$$\begin{aligned} \langle\phi\,|\,\hat{A}\,|\,\psi\rangle &= \langle\phi\,|\,\hat{I}\hat{A}\hat{I}\,|\,\psi\rangle = \langle\phi\,|\left(\sum_{n=1}^{\infty} |\,\phi_n\rangle\langle\phi_n\,|\right)\hat{A}\left(\sum_{m=1}^{\infty} |\,\phi_m\rangle\langle\phi_m\,|\right)|\,\psi\rangle \\ &= \sum_{nm}\langle\phi\,|\,\phi_n\rangle\langle\phi_n\,|\,\hat{A}\,|\,\phi_m\rangle\langle\phi_m\,|\,\psi\rangle \\ &= \sum_{nm} b_n^* A_{nm} a_m. \end{aligned} \tag{2.217}$$

This is a complex number; its matrix representation goes as follows:

$$\langle\phi\,|\,\hat{A}\,|\,\psi\rangle \longrightarrow (b_1^* \;\; b_2^* \;\; b_3^* \;\; \cdots)\begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ A_{31} & A_{32} & A_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \end{pmatrix}. \tag{2.218}$$

Remark
It is now easy to see explicitly why products of the type $|\,\psi\rangle\,|\,\phi\rangle$, $\langle\psi\,|\,\langle\phi\,|$, $\hat{A}\langle\psi\,|$, or $|\,\psi\rangle\hat{A}$ are forbidden. They cannot have matrix representations; they are nonsensical. For instance, $|\,\psi\rangle\,|\,\phi\rangle$ is represented by the product of two column matrices:

$$|\,\psi\rangle\,|\,\phi\rangle \longrightarrow \begin{pmatrix} \langle\phi_1\,|\,\psi\rangle \\ \langle\phi_2\,|\,\psi\rangle \\ \vdots \end{pmatrix}\begin{pmatrix} \langle\phi_1\,|\,\phi\rangle \\ \langle\phi_2\,|\,\phi\rangle \\ \vdots \end{pmatrix}. \tag{2.219}$$

This product is clearly not possible to perform, for the product of two matrices is possible only when the number of columns of the first is equal to the number of rows of the second; in (2.219) the first matrix has one single column and the second an infinite number of rows.

2.5.1.5 Properties of a Matrix A

- Real if $A = A^*$ or $A_{mn} = A^*_{mn}$
- Imaginary if $A = -A^*$ or $A_{mn} = -A^*_{mn}$
- Symmetric if $A = A^T$ or $A_{mn} = A_{nm}$
- Antisymmetric if $A = -A^T$ or $A_{mn} = -A_{nm}$ with $A_{mm} = 0$
- Hermitian if $A = A^\dagger$ or $A_{mn} = A^*_{nm}$
- Anti-Hermitian if $A = -A^\dagger$ or $A_{mn} = -A^*_{nm}$
- Orthogonal if $A^T = A^{-1}$ or $AA^T = I$ or $(AA^T)_{mn} = \delta_{mn}$
- Unitary if $A^\dagger = A^{-1}$ or $AA^\dagger = I$ or $(AA^\dagger)_{mn} = \delta_{mn}$

Example 2.15
Consider a matrix A (which represents an operator $\hat{A}$), a ket $|\,\psi\rangle$, and a bra $\langle\phi\,|$:

$$A = \begin{pmatrix} 5 & 3+2i & 3i \\ -i & 3i & 8 \\ 1-i & 1 & 4 \end{pmatrix}, \quad |\,\psi\rangle = \begin{pmatrix} -1+i \\ 3 \\ 2+3i \end{pmatrix}, \quad \langle\phi\,| = \begin{pmatrix} 6 & -i & 5 \end{pmatrix}.$$

(a) Calculate the quantities $A\,|\,\psi\rangle$, $\langle\phi\,|\,A$, $\langle\phi\,|\,A\,|\,\psi\rangle$, and $|\,\psi\rangle\langle\phi\,|$.
(b) Find the complex conjugate, the transpose, and the Hermitian conjugate of A, $|\,\psi\rangle$, and $\langle\phi\,|$.
(c) Calculate $\langle\phi\,|\,\psi\rangle$ and $\langle\psi\,|\,\phi\rangle$; are they equal? Comment on the differences between the complex conjugate, Hermitian conjugate, and transpose of kets and bras.

Solution
(a) The calculations are straightforward:

$$A\,|\,\psi\rangle = \begin{pmatrix} 5 & 3+2i & 3i \\ -i & 3i & 8 \\ 1-i & 1 & 4 \end{pmatrix} \begin{pmatrix} -1+i \\ 3 \\ 2+3i \end{pmatrix} = \begin{pmatrix} -5+17i \\ 17+34i \\ 11+14i \end{pmatrix}, \tag{2.220}$$

$$\langle\phi\,|\,A = \begin{pmatrix} 6 & -i & 5 \end{pmatrix} \begin{pmatrix} 5 & 3+2i & 3i \\ -i & 3i & 8 \\ 1-i & 1 & 4 \end{pmatrix} = \begin{pmatrix} 34-5i & 26+12i & 20+10i \end{pmatrix}, \tag{2.221}$$

$$\langle\phi\,|\,A\,|\,\psi\rangle = \begin{pmatrix} 6 & -i & 5 \end{pmatrix} \begin{pmatrix} 5 & 3+2i & 3i \\ -i & 3i & 8 \\ 1-i & 1 & 4 \end{pmatrix} \begin{pmatrix} -1+i \\ 3 \\ 2+3i \end{pmatrix} = 59+155i, \tag{2.222}$$

$$|\,\psi\rangle\langle\phi\,| = \begin{pmatrix} -1+i \\ 3 \\ 2+3i \end{pmatrix} \begin{pmatrix} 6 & -i & 5 \end{pmatrix} = \begin{pmatrix} -6+6i & 1+i & -5+5i \\ 18 & -3i & 15 \\ 12+18i & 3-2i & 10+15i \end{pmatrix}. \tag{2.223}$$

(b) To obtain the complex conjugate of A, $|\ \psi\rangle$, and $\langle\phi\ |$, we need simply to take the complex conjugate of their elements:

$$A^* = \begin{pmatrix} 5 & 3-2i & -3i \\ i & -3i & 8 \\ 1+i & 1 & 4 \end{pmatrix}, \quad |\ \psi\rangle^* = \begin{pmatrix} -1-i \\ 3 \\ 2-3i \end{pmatrix}, \quad \langle\phi\ |^* = \begin{pmatrix} 6 & i & 5 \end{pmatrix}. \tag{2.224}$$

For the transpose of A, $|\ \psi\rangle$, and $\langle\phi\ |$, we simply interchange columns with rows:

$$A^T = \begin{pmatrix} 5 & -i & 1-i \\ 3+2i & 3i & 1 \\ 3i & 8 & 4 \end{pmatrix}, \quad |\ \psi\rangle^T = \begin{pmatrix} -1+i & 3 & 2+3i \end{pmatrix}, \quad \langle\phi\ |^T = \begin{pmatrix} 6 \\ -i \\ 5 \end{pmatrix}. \tag{2.225}$$

The Hermitian conjugate can be obtained by taking the complex conjugates of the transpose expressions calculated above: $A^\dagger = (A^T)^*$, $|\ \psi\rangle^\dagger = \left(|\ \psi\rangle^T\right)^* = \langle\psi\ |$, $\langle\phi\ |^\dagger = \left(\langle\phi\ |^T\right)^* = |\ \phi\rangle$:

$$A^\dagger = \begin{pmatrix} 5 & i & 1+i \\ 3-2i & -3i & 1 \\ -3i & 8 & 4 \end{pmatrix}, \quad \langle\psi\ | = \begin{pmatrix} -1-i & 3 & 2-3i \end{pmatrix}, \quad |\ \phi\rangle = \begin{pmatrix} 6 \\ i \\ 5 \end{pmatrix}. \tag{2.226}$$

(c) Using the kets and bras above, we can easily calculate the needed scalar products:

$$\langle\phi\ |\ \psi\rangle = \begin{pmatrix} 6 & -i & 5 \end{pmatrix} \begin{pmatrix} -1+i \\ 3 \\ 2+3i \end{pmatrix} = 6(-1+i)+(-i)(3)+5(2+3i) = 4+18i, \tag{2.227}$$

$$\langle\psi\ |\ \phi\rangle = \begin{pmatrix} -1-i & 3 & 2-3i \end{pmatrix} \begin{pmatrix} 6 \\ i \\ 5 \end{pmatrix} = 6(-1-i)+(i)(3)+5(2-3i) = 4-18i. \tag{2.228}$$

We see that $\langle\phi\ |\ \psi\rangle$ and $\langle\psi\ |\ \phi\rangle$ are not equal; they are complex conjugates of each other:

$$\langle\psi\ |\ \phi\rangle = \langle\phi\ |\ \psi\rangle^* = 4-18i. \tag{2.229}$$

Remark
We should underscore the importance of the differences between $|\ \psi\rangle^*$, $|\ \psi\rangle^T$, and $|\ \psi\rangle^\dagger$. Most notably, we should note (from equations (2.224)–(2.226)) that $|\ \psi\rangle^*$ is a ket, while $|\ \psi\rangle^T$ and $|\ \psi\rangle^\dagger$ are bras. Additionally, we should note that $\langle\phi\ |^*$ is a bra, while $\langle\phi\ |^T$ and $\langle\phi\ |^\dagger$ are kets.

2.5.2 Change of Bases and Unitary Transformations

In a Euclidean space, a vector $\vec{A}$ may be represented by its components in different coordinate systems or in different bases. The transformation from one basis to the other is called a change of basis. The components of $\vec{A}$ in a given basis can be expressed in terms of the components of $\vec{A}$ in another basis by means of a *transformation* matrix.

Similarly, state vectors and operators of quantum mechanics may also be represented in different bases. In this section we are going to study how to transform from one basis to another. That is, knowing the components of kets, bras, and operators in a basis $\{|\ \phi_n\rangle\}$, how

does one determine the corresponding components in a different basis $\{|\ \phi'_n\rangle\}$? Assuming that $\{|\ \phi_n\rangle\}$ and $\{|\ \phi'_n\rangle\}$ are two different bases, we can expand each ket $|\ \phi_n\rangle$ of the old basis in terms of the new basis $\{|\ \phi'_n\rangle\}$ as follows:

$$|\ \phi_n\rangle\ =\ \left(\sum_m |\ \phi'_m\rangle\langle\phi'_m\ |\right)|\ \phi_n\rangle\ =\ \sum_m U_{mn}\ |\ \phi'_m\rangle, \tag{2.230}$$

where

$$U_{mn}\ =\ \langle\phi'_m\ |\ \phi_n\rangle. \tag{2.231}$$

The matrix U, providing the transformation from the old basis $\{|\ \phi_n\rangle\}$ to the new basis $\{|\ \phi'_n\rangle\}$, is given by

$$U = \begin{pmatrix} \langle\phi'_1\ |\ \phi_1\rangle & \langle\phi'_1\ |\ \phi_2\rangle & \langle\phi'_1\ |\ \phi_3\rangle \\ \langle\phi'_2\ |\ \phi_1\rangle & \langle\phi'_2\ |\ \phi_2\rangle & \langle\phi'_2\ |\ \phi_3\rangle \\ \langle\phi'_3\ |\ \phi_1\rangle & \langle\phi'_3\ |\ \phi_2\rangle & \langle\phi'_3\ |\ \phi_3\rangle \end{pmatrix}. \tag{2.232}$$

Example 2.16 (Unitarity of the transformation matrix)
Let U be a transformation matrix which connects two *complete* and *orthonormal* bases $\{|\ \phi_n\rangle\}$ and $\{|\ \phi'_n\rangle\}$. Show that U is *unitary*.

Solution
For this we need to prove that $\hat{U}\hat{U}^\dagger = \hat{I}$, which reduces to showing that $\langle\phi_m\ |\ \hat{U}\hat{U}^\dagger\ |\ \phi_n\rangle\ = \delta_{mn}$. This goes as follows:

$$\langle\phi_m\ |\ \hat{U}\hat{U}^\dagger\ |\ \phi_n\rangle\ =\ \langle\phi_m\ |\ \hat{U}\left(\sum_l |\ \phi_l\rangle\langle\phi_l\ |\right)\hat{U}^\dagger\ |\ \phi_n\rangle\ =\ \sum_l U_{ml}U^*_{nl}, \tag{2.233}$$

where $U_{ml}\ =\ \langle\phi_m\ |\ \hat{U}\ |\ \phi_l\rangle$ and $U^*_{nl}\ =\ \langle\phi_l\ |\ \hat{U}^\dagger\ |\ \phi_n\rangle\ =\ \langle\phi_n\ |\ \hat{U}\ |\ \phi_l\rangle^*$. According to (2.231), $U_{ml} = \langle\phi'_m\ |\ \phi_l\rangle$ and $U^*_{nl} = \langle\phi_l\ |\ \phi'_n\rangle$; we can thus rewrite (2.233) as

$$\sum_l U_{ml}U^*_{nl} = \sum_l \langle\phi'_m\ |\ \phi_l\rangle\langle\phi_l\ |\ \phi'_n\rangle\ =\ \langle\phi'_m\ |\ \phi'_n\rangle\ =\ \delta_{mn}. \tag{2.234}$$

Combining (2.233) and (2.234), we infer $\langle\phi_m\ |\ \hat{U}\hat{U}^\dagger\ |\ \phi_n\rangle\ =\ \delta_{mn}$, or $\hat{U}\hat{U}^\dagger = \hat{I}$.

2.5.2.1 Transformations of Kets, Bras, and Operators

The components $\langle\phi'_n\ |\ \psi\rangle$ of a state vector $|\ \psi\rangle$ in a new basis $\{|\ \phi'_n\rangle\}$ can be expressed in terms of the components $\langle\phi_n\ |\ \psi\rangle$ of $|\ \psi\rangle$ in an old basis $\{|\ \phi_n\rangle\}$ as follows:

$$\langle\phi'_m\ |\ \psi\rangle\ =\ \langle\phi'_m\ |\ \hat{I}\ |\ \psi\rangle\ =\ \langle\phi'_m\ |\left(\sum_n |\ \phi_n\rangle\langle\phi_n\ |\right)|\ \psi\rangle\ =\ \sum_n U_{mn}\langle\phi_n\ |\ \psi\rangle. \tag{2.235}$$

This relation, along with its complex conjugate, can be generalized into

$$|\ \psi_{new}\rangle\ =\ \hat{U}\ |\ \psi_{old}\rangle, \qquad \langle\psi_{new}\ | = \langle\psi_{old}\ |\ \hat{U}^\dagger. \tag{2.236}$$

Let us now examine how operators transform when we change from one basis to another. The matrix elements $A'_{mn} = \langle \phi'_m \mid \hat{A} \mid \phi'_n \rangle$ of an operator $\hat{A}$ in the new basis can be expressed in terms of the old matrix elements, $A_{jl} = \langle \phi_j \mid \hat{A} \mid \phi_l \rangle$, as follows:

$$A'_{mn} = \langle \phi'_m \mid \left(\sum_j \mid \phi_j \rangle\langle \phi_j \mid \right) \hat{A} \left(\sum_l \mid \phi_l \rangle\langle \phi_l \mid \right) \mid \phi'_n \rangle = \sum_{jl} U_{mj} A_{jl} U^*_{nl}; \tag{2.237}$$

that is,

$$\hat{A}_{new} = \hat{U} \hat{A}_{old} \hat{U}^\dagger \qquad \text{or} \qquad \hat{A}_{old} = \hat{U}^\dagger \hat{A}_{new} \hat{U}. \tag{2.238}$$

We may summarize the results of the change of basis in the following relations:

$$\boxed{\mid \psi_{new} \rangle = \hat{U} \mid \psi_{old} \rangle, \qquad \langle \psi_{new} \mid = \langle \psi_{old} \mid \hat{U}^\dagger, \qquad \hat{A}_{new} = \hat{U} \hat{A}_{old} \hat{U}^\dagger,} \tag{2.239}$$

or

$$\boxed{\mid \psi_{old} \rangle = \hat{U}^\dagger \mid \psi_{new} \rangle, \qquad \langle \psi_{old} \mid = \langle \psi_{new} \mid \hat{U}, \qquad \hat{A}_{old} = \hat{U}^\dagger \hat{A}_{new} \hat{U}.} \tag{2.240}$$

These relations are similar to the ones we derived when we studied unitary transformations; see (2.146) and (2.147).

Example 2.17
Show that the operator $\hat{U} = \sum_n \mid \phi'_n \rangle\langle \phi_n \mid$ satisfies all the properties discussed above.

Solution
First, note that $\hat{U}$ is unitary:

$$\hat{U}\hat{U}^\dagger = \sum_{nl} \mid \phi'_n \rangle\langle \phi_n \mid \phi_l \rangle\langle \phi'_l \mid = \sum_{nl} \mid \phi'_n \rangle\langle \phi'_l \mid \delta_{nl} = \sum_n \mid \phi'_n \rangle\langle \phi'_n \mid = \hat{I}. \tag{2.241}$$

Second, the action of $\hat{U}$ on a ket of the old basis gives the corresponding ket from the new basis:

$$\hat{U} \mid \phi_m \rangle = \sum_n \mid \phi'_n \rangle\langle \phi_n \mid \phi_m \rangle = \sum_n \mid \phi'_n \rangle \delta_{nm} = \mid \phi'_m \rangle. \tag{2.242}$$

We can also verify that the action $\hat{U}^\dagger$ on a ket of the new basis gives the corresponding ket from the old basis:

$$\hat{U}^\dagger \mid \phi'_m \rangle = \sum_l \mid \phi_l \rangle\langle \phi'_l \mid \phi'_m \rangle = \sum_l \mid \phi_l \rangle \delta_{lm} = \mid \phi_m \rangle. \tag{2.243}$$

How does a trace transform under unitary transformations? Using the cyclic property of the trace, $\mathrm{Tr}(\hat{A}\hat{B}\hat{C}) = \mathrm{Tr}(\hat{C}\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{C}\hat{A})$, we can ascertain that

$$\mathrm{Tr}(\hat{A}') = \mathrm{Tr}(\hat{U}\hat{A}\hat{U}^\dagger) = \mathrm{Tr}(\hat{U}^\dagger \hat{U} \hat{A}) = \mathrm{Tr}(\hat{A}), \tag{2.244}$$